Poisson Distribution Lab Activity

Poisson Distribution Lab

Tasks:

- Stay in your groups from data collection to work together on the analysis.
- Get the compiled class data into JASP - I recommend tossing them into an excel sheet and saving that as as CSV file so you always have access to the raw data.
- You may want to change your data type once you get it into JASP - check your columns!

1. Produce of a histogram of the raw data and describe the pattern. (Is it normally distributed? Skewed? Etc.)
    a. Hint: you can change the axes labels of a plot in JASP by clicking on the down arrow next to the bolded word right above the plot and going to "edit image". Remember to hit the enter key!
2. What is the mean and variance? Are they similar?
3. Produce a histogram of the *expected* number of quadrats per individuals (so, for the probability of getting x individuals, multiply that by the total number of quadrats sampled).
    b. This will be the part that takes you the longest.
    c. We sampled plots, and they had between 0 and X individuals of interest in them.
    d. You have the mean number of individuals in a plot, and the formula for a Poisson Random Variable.
    e. You will want to figure out the probability of getting 0 individuals in a plot, getting 1 individual in a plot, 2 individuals in a plot, up to X individuals in a plot.
    f. Given those probabilities, if we had Y plots, how many of those would we have expected to have 0 individuals in them? 1? 2? X?
    g. Create a new column in your JASP dataset with these numbers. (Just like your count column, but it's expected counts - if we expect 10 plots to have no individuals in them, there should be 10 zeroes in that column.)
4. Compare the two histograms. The one generated assuming a Poisson process suggests individuals are randomly distributed. Do you think they are? Why/why not?
    h. You can also look back at your expected quadrats per individuals vs the actual, if you would like.

Deliverables:

- A PDF of JASP results that shows the summary statistics, and the two histograms. (5 points total)
- A PDF (or other document, or typed into Blackboard) that answers the questions raised in parts 1, 2, and 4. (3 points total)

As you worked in groups, you may turn in the same document(s) - the same JASP results, and if you worked together discussing the results and the questions, the same answers to the questions as your group members. However, each individual in the group needs to turn in something so it can be graded in

Blackboard!

This is due on Thursday, 09/10 at 11:59 PM.